

# C5100-48XQ



## Specification

| Hardware                          | Description                                                                                                |                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Performance</b>                | • Switching Capacity: 1.44 Tbps                                                                            | • Forwarding rate: 1,071 Mpps                                                |
| <b>Interface</b>                  | • 48 port 1G/10G SFP+                                                                                      | • 6 port 40G                                                                 |
| <b>Memory</b>                     | • DRAM 4 GB                                                                                                | • Flash 8 GB                                                                 |
| <b>FAN</b>                        | • 4 Modular FANs                                                                                           | • Hot swappable                                                              |
| <b>Management</b>                 | • 1-port RJ-45 Console<br>• 1-port MGMT                                                                    | • 1-port USB                                                                 |
| <b>Power Supply</b>               | • Dual power supplies<br>• Power Consumption: 200W                                                         | • Modular power supply with hot-swappable<br>• Heat Dissipation: 682.4 BTU/h |
| <b>Environment</b>                | • Operating Temperature: 0~50°C                                                                            | • Operating Humidity: 10~90% (Non-condensing)                                |
| <b>Dimension</b>                  | • 442(W) x 560(D) x 44(H) mm                                                                               | • 1 RU                                                                       |
| Software                          | Description                                                                                                |                                                                              |
| <b>Virtualization</b>             | • VS: Virtual Stacking<br>• MAD: Multi-Active Detection<br>• MPLS VPN                                      | • Max 8 Units Stack<br>• Multi VRF                                           |
| <b>Entry</b>                      | • Max 288K MAC Addresses<br>• 4K Active VLAN                                                               | • Max 175K IPv4 Routes<br>• 9K Jumbo Frame                                   |
| <b>VLAN</b>                       | • MAC / Protocol / IP Subnet-Based<br>• 802.1Q<br>• Voice VLAN                                             | • QinQ<br>• Private VLAN / Super-VLAN<br>• GVRP                              |
| <b>STP / Ring Protocol and LP</b> | • Standard STP / RSTP / MSTP<br>• Loop Detection and Prevention<br>• BPDU Guard / BPDU Filter / Root Guard | • PVSTP+ Compatible protocol<br>• ERPS<br>• Link Flap Detection              |
| <b>Aggregation</b>                | • Static Link Aggregation<br>• Member per group: 32                                                        | • Dynamic Link Aggregation(LACP)                                             |

| Software                   | Description                                                     |                                                       |
|----------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| <b>Multicast</b>           | • IGMP Snooping(with Querier and Fast Leave)                    | • IGMP v1/v2/v3                                       |
|                            | • IGMP Filter                                                   | • PIM-DM/SM/SSM                                       |
| <b>IPv6</b>                | • IPv4/IPv6 Dual Stack                                          | • IPv6 Management                                     |
|                            | • IPv6 ACL                                                      | • MLD Snooping                                        |
|                            | • VRRPv3                                                        | • RIPng / OSPFv3 / BGP4+                              |
| <b>L3 Features</b>         | • Support full L3 features without additional license           |                                                       |
|                            | • Static / RIP / OSPF / ISIS / BGP                              | • PBR(Policy-Based Routing)                           |
|                            | • BFD                                                           | • IPsec encryption for routing protocol               |
|                            | • VRRP                                                          | • Gateway Keepalive                                   |
| <b>Security</b>            | • L2~L4-based ACL                                               | • Port ACL, VLAN ACL, Global ACL                      |
|                            | • Time-Based ACL                                                | • 802.1x Authentication                               |
|                            | • Port Security                                                 | • DHCP Snooping                                       |
|                            | • Dynamic ARP Inspection                                        | • IP Source Guard                                     |
|                            | • DoS/DDoS Protection                                           | • Anti-ARP Spoofing prevention                        |
|                            | • CPP(CPU Protection Policy)                                    | • MAC Address Limiting                                |
|                            | • Enforced Password Change Upon First Login                     | • Strong Password Policy and Encryption Mechanism     |
|                            | • User isolation after APT prevention system interoperability * | • Checking the required security configuration *      |
|                            | • 8 Queues                                                      | • Queue Scheduling: SP / WRR / SP+WRR                 |
|                            | • Classification and Marking with DSCP /802.1p                  | • Traffic Shaping and Policing                        |
| <b>QoS</b>                 | • Policy-Based QoS                                              | • Tail-drop and WRED(Weighted Random Early Detection) |
|                            | • Buffer Management                                             |                                                       |
|                            |                                                                 |                                                       |
| <b>SDN</b>                 | • IP mobility by VxLAN encapsulation                            | • EVPN                                                |
|                            | • Netconf                                                       |                                                       |
| <b>Reliability</b>         | • OS Image Redundancy                                           | • Configuration Redundancy                            |
|                            | • Hot Patch                                                     | • Hardware Watchdog                                   |
| <b>Management Features</b> | • Ping and Traceroute                                           | • Console, Telnet and SSHv2                           |
|                            | • SNMPv1/2c/3 and RMON                                          | • LLDP / LLDP-MED                                     |
|                            | • Mirroring / Remote Mirroring / VLAN Mirroring                 | • 1:1, N:N Mirroring                                  |

\* CoreEdge Smart Controller required.

| Software                   | Description                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------|
| <b>Management Features</b> | • NTP                                                                                   |
|                            | • DHCP Server/Relay/Client                                                              |
|                            | • AAA(Authentication, Authorization, Accounting)                                        |
|                            | • RADIUS & TACACS+                                                                      |
|                            | • Syslog                                                                                |
|                            | • Configuration Rollback                                                                |
|                            | • DDM(Digital Diagnostic Monitoring)                                                    |
|                            | • ULFD (Uni-Directional Link Fault Detection)                                           |
|                            | • Flow Monitoring                                                                       |
|                            | • RBAC(Role-Based Access Control)                                                       |
|                            | • RTR(Response Time Reporter) for quality checking                                      |
|                            | • BSM features for switch diagnostics                                                   |
| <b>Test Report</b>         | • Embedded Event Script                                                                 |
|                            | • FTP / SFTP                                                                            |
|                            | • CSS is a sophisticated suite of management tools that provides a simplified approach. |
|                            | • ZTP(Zero Touch Provisioning) *                                                        |
|                            | • Sanity Report for switch and terminal *                                               |
|                            | • Scheduled firmware upgrade *                                                          |
| <b>Certification</b>       | • Scheduled config backup *                                                             |
|                            | • Auto-Backup/Upgrade/Provisioning/Recovery *                                           |
|                            | • End Device Management *                                                               |
|                            | • Switch Shaped Panel *                                                                 |
|                            | • System Health Monitoring *                                                            |

\* CoreEdge Smart Controller required.

| Standards                                            | Description                                         |
|------------------------------------------------------|-----------------------------------------------------|
| <b>IEEE</b>                                          | • IEEE 802.3ae 10G BASE-X                           |
|                                                      | • IEEE 802.3z 1000BASE-X                            |
|                                                      | • IEEE 802.3ba 40GbE                                |
|                                                      | • IEEE 802.3x Flow Control                          |
|                                                      | • IEEE 802.3az EEE(Energy Efficient Ethernet)       |
|                                                      | • IEEE 802.1AX Link Aggregation                     |
|                                                      | • IEEE 802.3ad Link Aggregation Control Protocol    |
|                                                      | • IEEE 802.1ag CFM(Connectivity Fault Management)   |
|                                                      | • IEEE 802.3ah (Ethernet in the First Mile)         |
|                                                      | • IEEE 802.1w RSTP(Rapid Spanning Tree Protocol)    |
|                                                      | • IEEE 802.1D STP(Spanning Tree Protocol)           |
|                                                      | • IEEE 802.1x(Port-Based Network Access Control)    |
|                                                      | • IEEE 802.1s MSTP(Multiple Spanning Tree Protocol) |
|                                                      | • IEEE 802.1Q VLAN Tagging                          |
|                                                      | • IEEE 802.1ad Provider Bridges(Q-in-Q)             |
|                                                      | • IEEE 802.1AB LLDP(Link Layer Discovery Protocol)  |
| • IEEE 802.1v VLAN Classification by Protocol & Port |                                                     |

| Standards | Description                                                        |
|-----------|--------------------------------------------------------------------|
| RFC       | • RFC 768 UDP (User Datagram Protocol)                             |
|           | • RFC 791 IP (Internet Protocol)                                   |
|           | • RFC 792 ICMP (Internet Control Message Protocol)                 |
|           | • RFC 793 TCP (Transmission Control Protocol)                      |
|           | • RFC 826 ARP (Address Resolution Protocol)                        |
|           | • RFC 854/855 Telnet and Tenet options                             |
|           | • RFC 894 Core IP Implementation for Ethernet                      |
|           | • RFC 919 IP Subnet Broadcasting                                   |
|           | • RFC 950 & RFC 951 Internet Standard Subnetting Procedure & BootP |
|           | • RFC 1027 Proxy ARP                                               |
|           | • RFC 1035 DNS Client                                              |
|           | • RFC 1042 IP over IEEE 802 Networks                               |
|           | • RFC 1058 RIP (Routing Information Protocol)                      |
|           | • RFC 1071 Internet Checksum Calculation                           |
|           | • RFC 1122 Internet Host Requirements                              |
|           | • RFC 1157 SNMP                                                    |
|           | • RFC 1191 Path MTU Discovery                                      |
|           | • RFC 1213 MIB-II                                                  |
|           | • RFC 1239 Standard MIB                                            |
|           | • RFC 1350 TFTP (Trivial File Transfer Protocol)                   |
|           | • RFC 1591 DNS (Domain Name System)                                |
|           | • RFC 1724 RIPv2 MIB Extension                                     |
|           | • RFC 1812 Requirements for IPv4 Routers                           |
|           | • RFC 1918 IP Addressing                                           |
|           | • RFC 1981 Path MTU Discovery for IPv6                             |
|           | • RFC 2082 RIP-2 MD5 Authentication                                |
|           | • RFC 2131 DHCPv4 Client                                           |
|           | • RFC 2328 OSPF Version 2                                          |
|           | • RFC 2453 RIPv2                                                   |
|           | • RFC 2460 IPv6 Specification                                      |
|           | • RFC 2464 Transmission of IPv6 Packets over Ethernet Networks     |
|           | • RFC 2578 SMIv2                                                   |
|           | • RFC 2581 TCP Congestion Control                                  |
|           | • RFC 2711 IPv6 Router Alert Option                                |
|           | • RFC 2819 RMON MIB (groups 1,2,3 & 9)                             |
|           | • RFC 2863 Interfaces Group MIB                                    |
|           | • RFC 2865 RADIUS Authentication                                   |
|           | • RFC 2866 RADIUS Accounting                                       |
|           | • RFC 3021 Using 31-Bit Prefixes on IPv4 Point-to-Point Links      |
|           | • RFC 3046 DHCP Relay Agent Information Option                     |
|           | • RFC 3315 DHCPv6 Client                                           |
|           | • RFC 3411 SNMP Architecture                                       |
|           | • RFC 3414 USM for SNMPv3                                          |
|           | • RFC 3416 v2 of the Protocol Operations for the SNMP              |
|           | • RFC 3418 MIB for SNMP                                            |
|           | • RFC 3587 IPv6 Global Unicast Address Format                      |
|           | • RFC 3484 Default Address Selection for IPv6                      |
|           | • RFC 3596 DNS Extensions for IPv6                                 |
|           | • RFC 3646 DNS Configuration Options for DHCPv6                    |
|           | • RFC 3768 VRRP                                                    |
|           | • RFC 4007 IPv6 Scoped Address Architecture                        |
|           | • RFC 4022 TCP (MIB for the Transmission Control Protocol)         |
|           | • RFC 4113 MIB for the UDP (User Datagram Protocol)                |
|           | • RFC 4213 Transition Mechanisms for IPv6 Hosts & Routers          |
|           | • RFC 4291 IPv6 Addressing Architecture                            |
|           | • RFC 4292 IP Forwarding Table MIB                                 |
|           | • RFC 4293 MIB for the IP (Internet Protocol)                      |
|           | • RFC 4318 Definitions of Managed Objects for Bridges with RSTP    |
|           | • RFC 4330 SNTP version 4                                          |
|           | • RFC 4443 ICMPv6                                                  |
|           | • RFC 4861 Neighbor Discovery for IPv6                             |
|           | • RFC 5014 IPv6 Socket API for Source Address Selection            |
|           | • RFC 5095 Deprecation of Type 0 Routing Headers in IPv6           |
|           | • RFC 5424 Syslog Protocol                                         |
|           | • RFC 5905 NTP version 4                                           |
|           | • RFC 4607 Source-Specific Multicast for IP                        |
|           | • RFC 3973 PIM-DM                                                  |
|           | • RFC 7761 PIM-SM                                                  |
|           | • RFC 4271 BGP-4                                                   |
|           | • RFC 4760 MP-BGP                                                  |